

Building a Small Graph Benchmark for User-Level Stance Classification on a Polarized Reddit r/changemyview Thread

Shan Gao
McGill University
Canada
shan.gao5@mail.mcgill.ca

Abstract

This project studies whether a polarized online discussion can be turned into a useful graph benchmark for user-level stance classification. Using one Reddit /r/changemyview thread on abortion, we construct a user-reply graph, aggregate each user’s comments into a text representation, and manually assign stance labels from {*pro-choice*, *pro-life*, *neutral/mixed*}. We then compare text-only and graph-based models using TinyBERT features and repeated stratified evaluation. The resulting prototype contains 47 labeled users, which is too small for strong conclusions, but it is sufficient to demonstrate an benchmark creation pipeline and to provide a first comparison between standard text baselines and graph-based methods in this setting.

1 Introduction and Motivation

This project is a benchmark creation project. The goal is to build a small graph dataset for one polarized topic on social media, run baseline models on it, and test whether the graph actually contains useful information for the prediction task. Polarized discussions are a natural graph setting because users do not post independently: they reply, argue, and reinforce each other. This makes the problem a node classification task on an interaction graph, where the main question is whether user stance can be predicted better when both text and graph structure are used instead of text alone.

Reddit is practical source for a course-scale benchmark because the reply structure is explicit, comments are naturally organized into trees, and public discussion threads can be collected more easily using standard Reddit tooling. In particular, CMV threads are appealing because they are built around disagreement and argument, which makes stance annotation more meaningful than in generic discussion forums.

Conversational stance detection is relevant because stance is often not expressed in a direct or self-contained way. In many online discussions, users write short replies such as “exactly,” “that is not true,” or “I agree,” which can be difficult to interpret without knowing what earlier comment they are responding to. Prior work has emphasized that stance in conversations depends strongly on reply context rather than isolated text alone [6]. As a result, treating each comment or each user’s text in isolation may lose important information about agreement, disagreement, and the broader argumentative context. In polarized discussions, this context is especially important, because the same short phrase can support very different positions depending on the surrounding conversation.

This motivates representing the discussion as a graph. At the user level, each node corresponds to a user and each edge represents a reply interaction. Such a representation captures not only what a user says, but also who they respond to and how their position

is expressed through the structure of the conversation. Figure 1 illustrates this point with a small reply chain adapted from the original Reddit data. The example shows that a short response may reveal little on its own, but becomes much easier to interpret once the preceding replies are visible. This is precisely the kind of setting in which graph structure may provide useful signal beyond text features alone.

More broadly, this project is motivated by a benchmark question as much as by a modeling question. It is easy to assume that social media conversations should naturally benefit from graph-based learning, but this should be tested rather than taken for granted. A useful benchmark should therefore not only package the data into a graph format, but also compare graph-based models against simple text-only baselines. In this way, the project asks whether the reply graph is genuinely informative for stance classification, or whether most of the predictive signal already comes from the text written by each user.

2 Related Work

Social media platforms contain many discussions in which users express opinions on controversial topics. Analyzing these discussions can help identify patterns of public opinion on issues such as abortion, which is the focus of this project, and epidemic prevention [2]. An important benchmark in this area is SemEval-2016 [5], which framed stance detection as a supervised classification task. Our work differs in that it studies stance at the user level and incorporates the reply graph between users rather than relying only on isolated text. The main inspiration is [8], which models controversial online discussions as graph-structured data with users, posts, and reply relations, and studies prediction tasks on top of that graph. However, following the critique in [1], this project treats benchmark construction and benchmark validation as separate goals. The concern is not only how to build a graph benchmark but also whether the chosen graph is actually a meaningful abstraction for the task. As the position paper argues, graph benchmarks should justify why a relational structure is useful for prediction and should include non-graph baselines to test whether the graph carries task-relevant signal.

Related work for stance classification, such as [3], shows that reply threads naturally form a graph structure, but much of this literature focuses on sequence- or feature-based models rather than small user-level graph benchmarks. In addition, reply graphs in controversial discussions are likely to contain substantial heterophily because users often reply to people they disagree with. This matters because standard GCNs are known to rely most heavily on homophily assumptions [9]. For that reason, this project also includes a GraphSAGE-style baseline, which is often considered more

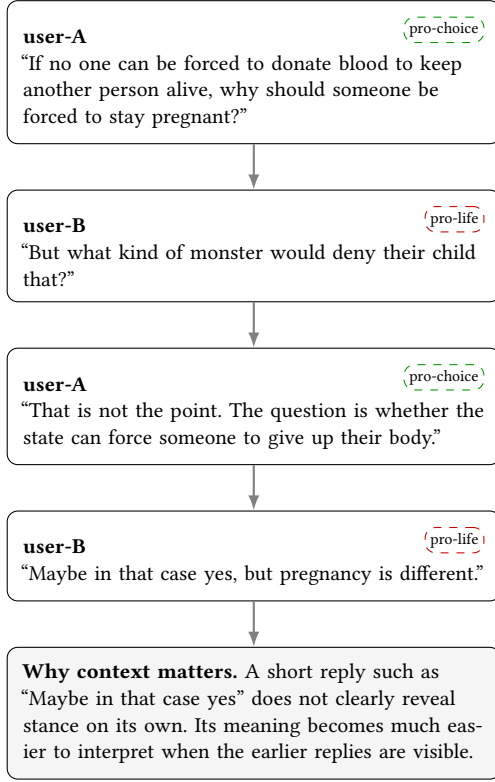


Figure 1: Adapted from a real reply chain in the original Reddit thread. Short replies can be hard to interpret in isolation, and earlier replies provide the context needed to infer stance.

robust in settings where neighborhood smoothing may be less appropriate [4].

3 Problem Definition

We define a directed graph $G = (V, E)$ where each node $v \in V$ is a user and an edge $(u, v) \in E$ means that user u replied to user v . Each user has a stance label

$$y_v \in \{\text{pro-choice, pro-life, neutral/mixed}\}.$$

Notice that these labels are adapted to the specific dataset that we are considering. The input is: (1) a polarized discussion thread, (2) the user reply graph, (3) text written by each user, and (4) manual stance labels. The output is a predicted stance label for each user. The central research question is whether user interactions contain task-relevant information beyond what can already be inferred from aggregated user text.

4 Dataset Description

4.1 Reddit thread selection

To collect authentic social media interaction data, we used Reddit, one of the largest online discussion platforms. Because our request for API access was never answered, we relied on historical Reddit dumps instead. Since searching through large Reddit dumps repeatedly is expensive, we selected one thread from /r/changemyview

and manually checked that it was suitable for the project. In particular, we verified that the thread was relevant to the topic, had enough comments and engagement, and contained sufficiently detailed discussion to support manual annotation.

4.2 Data Preprocessing

We performed the following preprocessing steps:

- Collect the full Reddit thread, including top-level comments, nested replies, usernames, parent comment information, and comment text.
- Remove deleted or removed comments, since these do not provide usable text for stance annotation.
- Normalize user identifiers so that each author is represented consistently across the thread.
- Extract the reply structure by matching each comment to its parent comment and, when possible, to the parent author.
- Keep only comments written by users that can be linked to a valid author name and a valid text field.
- Group all comments written by the same user into a single collection, so that annotation can be done at the user level rather than the individual comment level.
- Concatenate or review each user’s comments together in chronological or thread order to preserve conversational context during annotation.
- Remove exact duplicates or obvious formatting artifacts introduced during scraping or export.
- Standardize whitespace, line breaks, and quotation formatting to make the text easier to read during manual annotation.
- Store the cleaned data in a table containing at least the user ID, comment text, parent user or parent comment, and thread position.
- Create an annotation file in which each row corresponds to one user, together with their aggregated comments and an empty field for the stance label.

4.3 Selected threads

The dataset is built from a Reddit /r/changemyview (CMV) thread on abortion, chosen because the discussion is strongly stance-oriented and contains many direct replies between users with opposing views. At the raw comment level, the thread forms a reply forest. This is converted into a user-level interaction graph by aggregating all comments written by the same user and adding an edge from user u to user v whenever u replies to v at least once.

The dataset contains 47 labeled users. The class distribution is 25 pro-choice users, 11 pro-life users, and 11 neutral/mixed users. At the graph level, the directed NetworkX graph contains 47 nodes and 78 directed edges. For the GCN baseline, this graph is symmetrized and converted into a normalized adjacency matrix with self-loops. For the GraphSAGE baseline, the graph is represented as a PyTorch Geometric edge list with both directions included, yielding an edge index of size 2×156 .

Each node is paired with an aggregated text field obtained by concatenating all comments written by that user in the thread. These texts are used to build node features for both text-only and graph-based baselines.

4.4 Annotation Process

Due to time constraints for the course project, only a very small user-level graph was selected for annotation. Rather than attempting to label a larger dataset, the goal was to produce a small prototype benchmark with labels that were as reliable as possible within the available time.

The annotation was carried out manually at the user level. For each user, all of their comments in the selected Reddit CMV thread were read together and used as evidence for assigning a single stance label. The label set was

$$y_v \in \{\text{pro-choice, pro-life, neutral/mixed}\}.$$

A user was labeled **pro-choice** if their comments consistently supported abortion access, **pro-life** if their comments consistently opposed it, and **neutral/mixed** if their position was unclear, inconsistent, or primarily descriptive.

The initial annotation was performed by the author by reading the full reply context and aggregating each user's comments before assigning a label. The labels were then independently checked by the author's labmate as a verification step. Out of the 47 labeled users, 45 labels agreed between the two annotators. This corresponds to a raw agreement rate of approximately

$$\frac{45}{47} \approx 95.7\%.$$

The two disagreements were resolved manually through discussion and a final label was assigned in each case.

Although this is only a small-scale verification procedure and not a full formal annotation study, it provides some evidence that the stance labels are reasonably consistent. At the same time, the small size of the dataset remains an important limitation of the benchmark, and the annotation process should be viewed as a course-project level prototype rather than a finalized large-scale dataset.

5 Methodology

The final pipeline consists of four stages: data preprocessing, graph construction, feature extraction, and model evaluation.

- **Data preprocessing.** The annotated CSV files are loaded and merged so that stance labels are aligned at the user level. Rows missing a user identifier or stance label are removed. Author and parent-author fields are normalized in the comment table before graph construction. Refer to section 4 for more details on preprocessing.
- **Graph construction.** A directed user-reply graph is built from the comment table. Each node corresponds to one labeled user with a valid user_id and stance. The NetworkX graph stores the annotated stance for each user and the aggregated text. A directed edge (u, v) is added when user u replies to user v . This graph is also weighted as the same pair of users can interact many times, where the weight of an edge (u, v) is the number of times user u replies to user v . The resulting graph has 47 nodes and 78 directed edges. For the GCN baseline, the adjacency is symmetrized, self-loops are added, and the standard normalized matrix $D^{-1/2}AD^{-1/2}$ is used.

- **Feature extraction.** For each labeled user v , we take the aggregated text field `all_text_v`, tokenize it with the TinyBERT WordPiece tokenizer, and truncate or pad the sequence to a maximum length of 256 tokens. Along with the token IDs, the tokenizer returns an attention mask, where a value of 1 marks a real token and a value of 0 marks padding added only to equalize sequence lengths within a batch. The tokenized sequence is then passed through the pretrained model TinyBERT, which produces contextual hidden states $\mathbf{h}_{v,t} \in \mathbb{R}^{312}$ at each token position t . To obtain a single embedding for user v , we average only over the positions corresponding to real tokens, excluding padded positions using the attention mask:

$$\mathbf{x}_v = \frac{1}{|T_v|} \sum_{t \in T_v} \mathbf{h}_{v,t},$$

where $T_v = \{t : \text{attention_mask}_{v,t} = 1\}$. In other words, the attention mask is used to ensure that the pooled representation is computed only from actual text and is not distorted by padding tokens. These user embeddings are cached and aligned with the node ordering, so every node in the constructed graph is represented by the same 312-dimensional feature vector $\mathbf{x}_v$, which is then used as input to the text baselines, the GCN, and GraphSAGE.

- **Evaluation.** All reported results use repeated stratified 60%/20%/20% train/validation/test splits over 10 random seeds. In each split, this corresponds to 28 training users, 9 validation users, and 10 test users. Stratification is used to preserve the label proportions of the three stance classes as closely as possible across the train, validation, and test sets. This is especially important here because the dataset is small and imbalanced. Without stratification, a split could easily contain too few examples from one of the smaller classes, making the results unstable and hard to interpret.

For each split, we report both accuracy and macro-F1. This is because accuracy alone can be misleading because a model that predicts **pro-choice** too often may still obtain a reasonable score simply because that class is the largest.

For this reason, the primary evaluation metric is macro-F1. For each stance class $k \in \{\text{pro-choice, pro-life, neutral/mixed}\}$, let

$$\text{Precision}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}, \quad \text{Recall}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k},$$

where:

- TP_k is the number of users whose true stance is k and whose predicted stance is also k ,
- FP_k is the number of users predicted as k even though their true stance is different,
- FN_k is the number of users whose true stance is k but were predicted as one of the other two classes.

The F1 score for class k is then

$$\text{F1}_k = \frac{2 \text{Precision}_k \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}.$$

In this benchmark, this means that we compute one F1 score for **pro-choice**, one for **pro-life**, and one for **neutral/mixed**. The macro-F1 score is the unweighted average

of these three classwise F1 scores:

$$\text{Macro-F1} = \frac{1}{3} (\text{F1}_{\text{pro-choice}} + \text{F1}_{\text{pro-life}} + \text{F1}_{\text{neutral/mixed}}).$$

Macro-F1 is especially appropriate for this task because it gives equal importance to each stance class. As a result, a model must perform reasonably well not only on the larger **pro-choice** class, but also on the smaller **pro-life** and **neutral/mixed** classes.

Finally, because the graph contains only 47 labeled users, performance can vary substantially depending on the particular train/validation/test partition. To reduce the dependence on any single split, we repeat the stratified splitting procedure over 10 random seeds and report the mean and standard deviation of each metric across runs. If $m^{(1)}, \dots, m^{(S)}$ are the metric values over $S = 10$ splits, then the reported mean is

$$\bar{m} = \frac{1}{S} \sum_{s=1}^S m^{(s)},$$

and the corresponding standard deviation is

$$\sigma_m = \sqrt{\frac{1}{S} \sum_{s=1}^S (m^{(s)} - \bar{m})^2}.$$

Thus, the mean summarizes average performance for predicting **pro-choice**, **pro-life**, and **neutral/mixed** users across repeated stratified splits, while the standard deviation measures how sensitive the results are to the particular partition of this small dataset.

6 Experiment Setup

All models use BERT-based contextual embeddings of the aggregated user comments as input features. We evaluate four baselines:

- **Majority:** predict the majority class from the training set for all test users.
- **Text Logistic Regression:** multinomial logistic regression on the BERT text embeddings only.
- **Text MLP:** a small multilayer perceptron on the same BERT text embeddings only.
- **GCN:** a two-layer graph convolutional network using the BERT text embeddings together with the reply graph.
- **GraphSAGE:** a GraphSAGE model using the same BERT text embeddings together with the reply graph.

To elaborate on why GraphSAGE might be more appropriate for heterophilous datasets, GraphSAGE updates a node by first computing a separate neighbor summary

$$h_{N(v)}^k \leftarrow \text{AGGREGATE}_k(\{h_u^{k-1}, \forall u \in N(v)\})$$

and then combining it with the node's own representation

$$h_v^k \leftarrow \sigma(\mathbf{W}^k \cdot \text{CONCAT}(h_v^{k-1}, h_{N(v)}^k))$$

which can give it an edge over GCN on heterophilous graphs because it does not immediately average a node with neighbors that may have different labels. All models are evaluated using the same repeated stratified split procedure described above. The GCN uses early stopping based on validation macro-F1.

7 Visualization of the Graph

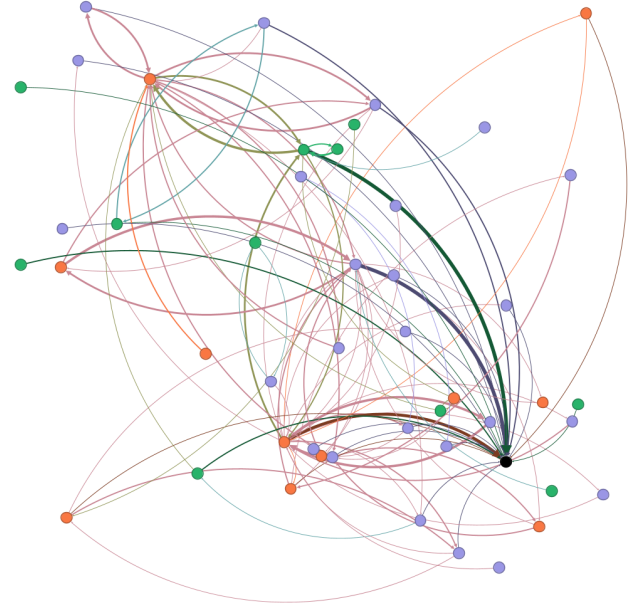


Figure 2: Gephi visualization of the user-level reply graph for the prototype Reddit CMV thread. Nodes represent users and directed edges represent reply interactions. Node color indicates stance label: purple for pro-choice, orange for pro-life, and green for neutral/mixed.

8 Results

Table 1 reports the mean and standard deviation over 10 repeated stratified splits for the baselines using TinyBERT text features, together with the GraphSAGE model using the same 312-dimensional node features.

Table 1: Results over 10 repeated stratified splits using TinyBERT-based text features.

Model	Macro-F1 (mean $\pm$ std)	Accuracy (mean $\pm$ std)
majority	0.2222 $\pm$ 0.0000	0.50 $\pm$ 0.0000
text_logreg_tinybert	0.3029 $\pm$ 0.1341	0.37 $\pm$ 0.1160
text_mlp_tinybert	0.2888 $\pm$ 0.1303	0.38 $\pm$ 0.1135
gcn_tinybert	0.2398 $\pm$ 0.0885	0.43 $\pm$ 0.1252
graphsage_tinybert	0.2489 $\pm$ 0.1074	0.35 $\pm$ 0.1179

Using TinyBERT-based text features, the text-only logistic regression model achieved the highest mean macro-F1, with 0.3029 $\pm$ 0.1341. The text MLP followed closely with a mean macro-F1 of 0.2888 $\pm$ 0.1303. Among the graph-based models, GCN with TinyBERT features obtained a mean macro-F1 of 0.2398 $\pm$ 0.0885, while GraphSAGE with TinyBERT features obtained 0.2489 $\pm$ 0.1074.

In terms of accuracy, the majority baseline achieved the highest mean value at 0.50 $\pm$ 0.0000. Among the learned models, GCN with

TinyBERT features obtained the highest mean accuracy, with 0.43 ± 0.1252 . The text MLP and text logistic regression models achieved mean accuracies of 0.38 ± 0.1135 and 0.37 ± 0.1160 , respectively, while GraphSAGE with TinyBERT features achieved 0.35 ± 0.1179 .

Across all models, the standard deviations show noticeable variation across the 10 random splits, both for macro-F1 and for accuracy.

9 Evaluation

The results should be read carefully because the dataset is very small. With only 47 labeled users, each test split contains very few examples, so even one or two predictions can noticeably change the final scores. This means that the reported numbers are useful for understanding general trends, but they are not strong enough to support broad claims.

Looking first at accuracy, the majority baseline performs better than all of the learned models. This happens because the dataset is imbalanced: the **pro-choice** class is the largest class, so a model that predicts the majority label very often can still obtain a relatively high accuracy. In other words, the majority baseline benefits from the label distribution of the dataset, even though it does not actually learn to distinguish between the three stance classes.

Among the non-majority baselines, the GCN achieves the best accuracy. This suggests that the reply graph may contain useful information for predicting user stance. One possible reason is that the graph may contain some degree of *homophily*, meaning that users connected by replies may sometimes share the same stance label. In that case, a graph-based model such as GCN can benefit from combining text features with information from neighboring nodes.

At the same time, the advantage is still limited, and the overall results remain unstable across splits. This means that the current benchmark is too small to determine with confidence how much of the GCN's performance comes from real graph signal and how much comes from variation in the train/test partition. A larger dataset would be needed to test this more reliably.

The results therefore suggest two things. First, raw accuracy alone can be misleading in this task, because it favors the majority baseline under class imbalance. Second, among the learned models, the GCN appears to make the best use of the available information for overall classification accuracy, which provides some evidence that the graph structure may be helpful in this benchmark.

Looking next at macro-F1, the picture is different. All learned models outperform the majority baseline, which is encouraging. This is important because macro-F1 gives equal weight to the **pro-choice**, **pro-life**, and **neutral/mixed** classes, so it is a better measure of balanced performance than raw accuracy. The fact that every trained model improves on the majority baseline suggests that the models are learning at least some signal beyond simply predicting the largest class.

At the same time, the simpler text-based baselines, namely logistic regression and MLP, achieve higher macro-F1 than the more complex graph-based models, GCN and GraphSAGE. This may suggest that, in the current benchmark, the reply graph does not provide enough additional signal to improve balanced stance classification beyond what is already available in the text features. In other words, for this small dataset, the graph structure may be too

weak, too noisy, or too limited to consistently help with prediction across all three classes.

However, it is also worth noting that GraphSAGE performs better than GCN in macro-F1. One possible explanation is that this task may involve some degree of *heterophily*, where users often reply to people they disagree with rather than people who share the same stance. In that setting, a standard GCN may be less suitable because it tends to work best when neighboring nodes are similar. GraphSAGE, while still a graph-based model, may be somewhat more flexible in handling this kind of structure. This makes the comparison between GraphSAGE and GCN meaningful: even though GCN achieves better accuracy, the stronger macro-F1 of GraphSAGE may indicate that heterophily is relevant for balanced stance prediction in this discussion graph.

10 Discussion

The main point of this project is that the dataset is too small to conclude whether it is really appropriate as a graph benchmark. With only 47 labeled users, it is also too small to conclude whether the graph structure truly provides enough signal for a graph-based approach to outperform more standard stance classification methods based only on text. The current results are therefore very preliminary, and they should be interpreted with caution.

At the same time, we hope this project still conveys that the process of benchmark creation has been carried out. A full pipeline was built: a relevant Reddit thread was selected, a user-level reply graph was constructed, stance labels were assigned, node features were extracted, and both text-only and graph-based baselines were evaluated. In that sense, the project shows that this kind of benchmark is feasible.

More broadly, there is still a lot of potential in this direction. Access to very large amounts of unprocessed Reddit data already exists, so the main bottleneck is not the availability of raw data, but the work needed to turn that data into a properly annotated and evaluated benchmark. For that reason, the current work should be viewed as a first step rather than a finished benchmark.

10.1 Limitations

One important limitation is that the data annotation process is not done correctly in the full benchmark sense. Many annotators are needed, together with explicit annotation guidelines. This is because annotation can be made more robust when several annotators vote on the correct label instead of relying on only one or two people. Important missing steps include a pilot annotation round and the computation of kappa statistics [7] as a measure of inter-annotator agreement.

The main limitation, however, is the size of the dataset. Since the goal is to test graph learning methods, the current dataset is nowhere near the size required for such a task. With such a small graph, it is very difficult to tell whether the observed results reflect real properties of the benchmark or are simply artifacts of the small sample.

10.2 Possible Future Work

Future work can include a better annotation process, more data, and more annotators. A stronger annotation pipeline would include

clear written guidelines, a pilot round to refine the labels, multiple annotators per user, and agreement statistics such as Cohen's kappa. On the data side, the most natural next step is to expand beyond a single thread and annotate a much larger collection of Reddit discussions. This would make it possible to test graph learning methods more seriously and to study more carefully whether the reply graph adds useful signal beyond text alone.

11 Conclusion

This project studied whether a polarized Reddit reply thread can be turned into a small graph benchmark for user-level stance classification, where the goal is to predict whether a user is **pro-choice**, **pro-life**, or **neutral/mixed**. The approach was to build a user-level reply graph from one */r/changemyview* thread, manually annotate the users, generate TinyBERT text features, and evaluate both text-only and graph-based baselines on repeated stratified splits. The main result is that the full benchmark creation pipeline was completed, but the final dataset of 47 labeled users is too small to determine whether the graph structure reliably adds useful signal beyond text alone. In the experiments, simple text baselines achieved the strongest macro-F1, while GCN achieved the best accuracy among the learned models, giving only weak and preliminary evidence that the graph may help. Overall, the work should be viewed as a proof of concept with clear potential for expansion to a much larger and better-annotated Reddit benchmark.

References

- [1] Maya Bechler-Speicher, Ben Finkelshtein, Fabrizio Frasca, Luis Müller, Jan Tönshoff, Antoine Siraudin, Viktor Zaverkin, Michael M. Bronstein, Mathias Niepert, Bryan Perozzi, Mikhail Galkin, and Christopher Morris. 2025. Position: Graph Learning Will Lose Relevance Due To Poor Benchmarks. In *Forty-second International Conference on Machine Learning Position Paper Track*. <https://openreview.net/forum?id=nDFpl2lhoH>
- [2] Kyle Glandt, Sarthak Khanal, Yingjie Li, Doina Caragea, and Cornelia Caragea. 2021. Stance Detection in COVID-19 Tweets. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (Eds.). Association for Computational Linguistics, Online, 1596–1611. doi:10.18653/v1/2021.acl-long.127
- [3] Genevieve Gorrell, Kalina Bontcheva, Leon Derczynski, Elena Kochkina, Maria Liakata, and Arkaitz Zubiaga. 2018. RumourEval 2019: Determining Rumour Veracity and Support for Rumours. arXiv:1809.06683 [cs.CL] <https://arxiv.org/abs/1809.06683>
- [4] William L. Hamilton, Rex Ying, and Jure Leskovec. 2018. Inductive Representation Learning on Large Graphs. arXiv:1706.02216 [cs.SI] <https://arxiv.org/abs/1706.02216>
- [5] Saif Mohammad, Svetlana Kiritchenko, Parinaz Sobhani, Xiaodan Zhu, and Colin Cherry. 2016. SemEval-2016 Task 6: Detecting Stance in Tweets. In *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, Steven Bethard, Marine Carpuat, Daniel Cer, David Jurgens, Preslav Nakov, and Torsten Zesch (Eds.). Association for Computational Linguistics, San Diego, California, 31–41. doi:10.18653/v1/S16-1003
- [6] Ramon Villa-Cox, Sumeet Kumar, Matthew Babcock, and Kathleen M. Carley. 2020. Stance in Replies and Quotes (SRQ): A New Dataset For Learning Stance in Twitter Conversations. arXiv:2006.00691 [cs.CL] <https://arxiv.org/abs/2006.00691>
- [7] Wikipedia contributors. 2025. Cohen's kappa — Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Cohen%27s_kappa&oldid=1329526122 [Online; accessed 12-April-2026].
- [8] Yilun Zheng, Sha Li, Fangkun Wu, Yang Ziyi, Lin Hongchao, Zhichao Hu, Cai Xinjun, Ziming Wang, Jinxuan Chen, Sitao Luan, Jiahao Xu, and Lihui Chen. 2025. FanChuan: A Multilingual and Graph-Structured Benchmark For Parody Detection and Analysis. arXiv:2502.16503 [cs.CL] <https://arxiv.org/abs/2502.16503>
- [9] Jiong Zhu, Yujun Yan, Lingxiao Zhao, Mark Heimann, Leman Akoglu, and Danai Koutra. 2020. Beyond Homophily in Graph Neural Networks: Current Limitations and Effective Designs. arXiv:2006.11468 [cs.LG] <https://arxiv.org/abs/2006.11468>